HR Attrition

AI Insight Generator

Prompt Engineering · Fine-Tuning · Interactive App

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Project Overview

The Problem

HR teams deal with noisy, unstructured attrition data but rarely have tools that translate that data into clear narratives fast. The goal of this project was to test whether a generative AI model and eventually a fine-tuned one could fill that gap reliably.

Milestone 2

Prompt Experiments

Tested 5 prompt types against the Anthropic API: summaries, recommendations, multi-dept analysis, scenario planning, and audience adaptation.

Milestone 3

Fine-Tuning

Built a 20-example JSONL training set, uploaded to OpenAI, ran a supervised fine-tuning job on gpt-3.5-turbo-0125, and compared outputs.

Milestone 4

Interactive App

Wrapped the fine-tuned model in an ipywidgets UI inside Jupyter - dropdown mode selector, text input, and a Generate button.

Milestone 2 - Prompt Experiments

01

Basic Summary

✓ Accurately parsed attrition data and surfaced key risk groups without hallucinating numbers.

02

Recommendations

✓ Returned 3 specific, actionable retention suggestions tied directly to the stated exit reasons.

03

Multi-Dept Analysis

✓ Correctly ranked Engineering and Sales as highest risk and suggested one company-wide action.

04

Scenario Planning

✓ Generated clearly differentiated conservative / moderate / aggressive staffing scenarios.

05

Audience Adaptation

✓ Adjusted tone and detail level for exec vs. manager, same insight, two different voices.

Milestone 3 - Fine-Tuning

Step 1

Build Dataset

20 JSONL examples
(summaries, recs, scenarios, audience, multi-dept)

Step 2

Upload File

POST /files
purpose: fine-tune
file-XGu3P7Yz13Vx...

Step 3

Create Job

POST /fine_tuning/jobs
ftjob-y48mpGso...
100 training steps

Step 4

Evaluate

Base vs. fine-tuned
Same prompt, side-by-side comparison

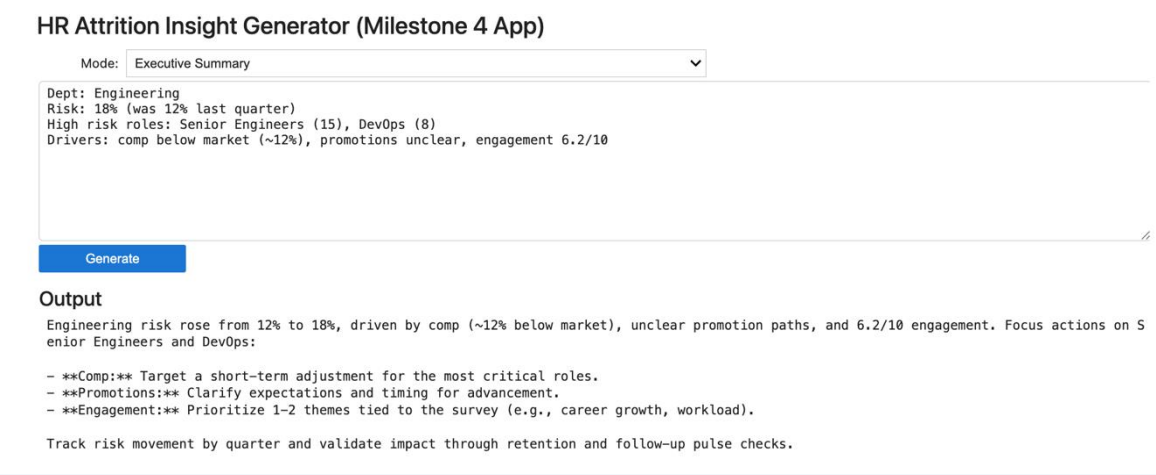
Model Comparison (same prompt)

Base model: *"...addressing these issues is crucial to retaining talent in the department."*

Fine-tuned: *"A focused retention plan should prioritize targeted raises for the at-risk groups and clearer career development support."*

Fine-tuned model drops the filler closing and goes straight to the next action.

Milestone 4 - Interactive App



5 Analysis Modes

Summary, Recommendations, Multi-Dept, Scenario Planning, Audience Adaptation

Token-Safe

200 max output tokens, temp=0.2, 3-second cooldown prevents accidental API spam

Fine-Tuned Model

ft:gpt-3.5-turbo-0125:personal:hr-attrition-m3 - consistent HR-specific tone and structure

Key Takeaways

01 Base model handled HR reasoning well out of the box - prompt engineering alone got usable results across all 5 experiment types.

02 Fine-tuning isn't about teaching new facts. It's about locking in tone, format, and structure so the model stops improvising on things that should be consistent.

03 The biggest operational lesson: quota management is a real constraint. Running a job twice to document it cost more tokens than the first run.

04 Wrapping the model in a simple UI (even inside Jupyter) changes how useful it feels - hiding the API details and giving a non-technical user a button to click is meaningful.